Data Analysis Lab – 20/08/2026

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Practical: GENOME ASSEMBLY AND ANNOTATION PIPELINE

OBJECTIVE

To construct a *de novo* genome assembly from paired-end short-read sequencing data, refine the assembly via reference-guided scaffolding, evaluate quality and gene content completeness, perform *ab initio* gene predictions, and identify the source organism using remote homology searching.

PIPELINE AND WORKFLOW EXECUTION

1. Environment Initialization & Dependency Setup

All bioinformatics modules necessary for downstream execution were deployed within a dedicated Conda environment:

Bash

```
pixi add --channel conda-forge --channel bioconda velvet quast augustus gffread ragtag busco seqtk
```

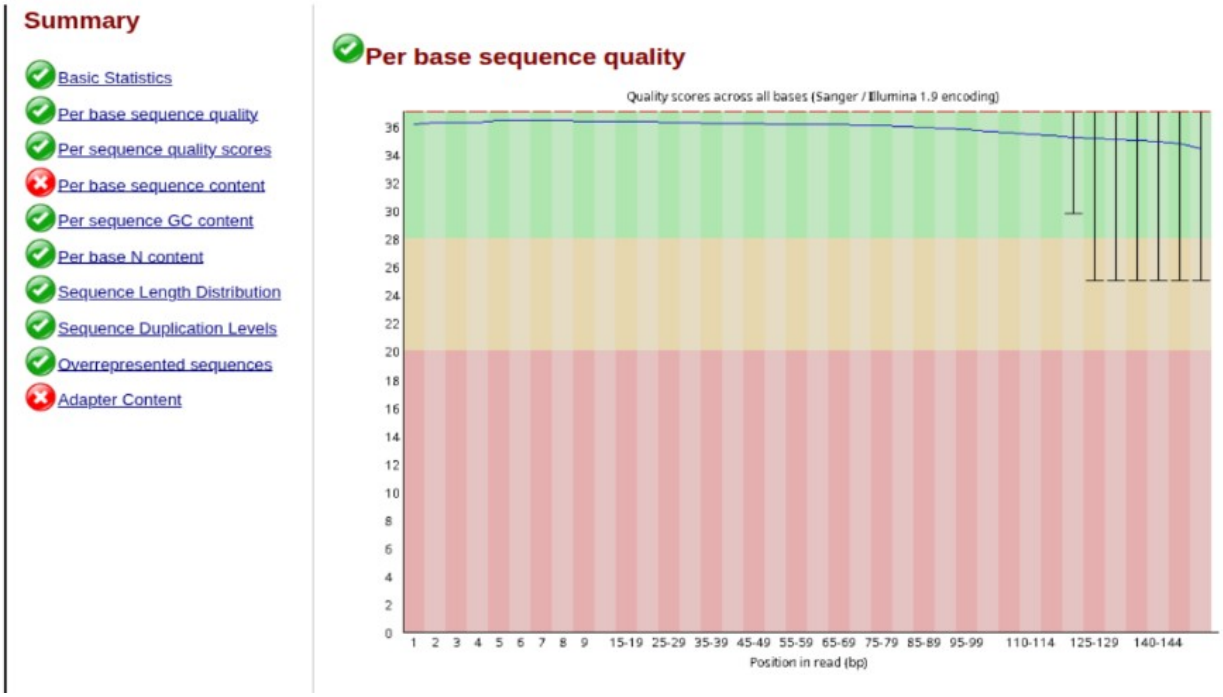
2. Read Quality Assessment

Initial diagnostic evaluation of read quality across paired-end FASTQ datasets (SR1.fastq and SR2.fastq) was carried out using FastQC:

Bash

```
pixi run fastqc SR1.fastq SR2.fastq
```

SR1.fastqc result:



- **Result:** High mean Phred quality scores (> 32) across sequence lengths. No significant adapter contamination or tile-level anomalies were identified.

3. De Novo Assembly Setup (Velvet)

De Bruijn graph-based assembly was performed using Velvet. The dataset was first indexed and hashed using a default k-mer size of k=31

Bash

```
velveth Assembly_step1 31 -shortPaired -fastq -separate SR1.fastq SR2.fastq
```

The graph was then resolved to construct contigs with a minimum node coverage cutoff of 5x:

Bash

```
velvetg Assembly_step1 -cov_cutoff 5 -unused_reads yes
```

4. Initial Draft Contig Metrics

To quantify total initial contigs generated:

Bash

grep -c ">" Assembly_step1/contigs.fa

```
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$ grep -c ">" Assembly_step1/contigs.fa
1688
```

- **Result:** total contigs = 1,688.

The raw contigs were evaluated without a reference using QUAST:

Bash

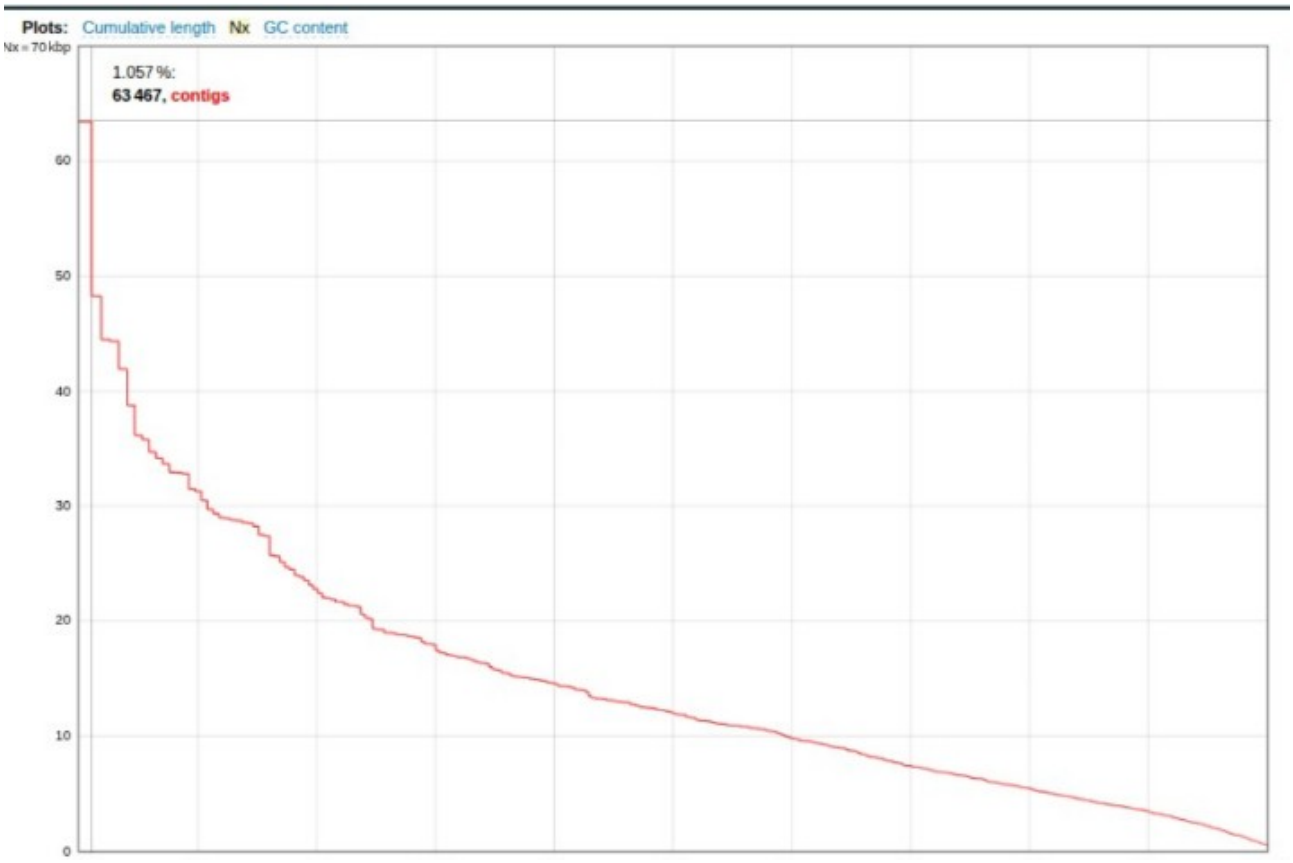
quast.py -o quast_contigs Assembly_step1/contigs.fa

20 August 2026, Thursday, 10:25:42

View in Icarus contig browser

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include

Statistics without reference	contigs
# contigs	878
# contigs (>= 0 bp)	1686
# contigs (>= 1000 bp)	757
# contigs (>= 5000 bp)	393
# contigs (>= 10000 bp)	203
# contigs (>= 25000 bp)	31
# contigs (>= 50000 bp)	1
Largest contig	63 467
Total length	6 002 093
Total length (>= 0 bp)	6 123 510
Total length (>= 1000 bp)	5 915 805
Total length (>= 5000 bp)	4 917 580
Total length (>= 10000 bp)	3 568 105
Total length (>= 25000 bp)	1 039 218
Total length (>= 50000 bp)	63 467
N50	12 014
N90	3422
auN	14 930
L50	152
L90	510
GC (%)	42.97
Per base quality	
# N's per 100 kbp	0
# N's	0



5. Structural Correction and Reference-Guided Scaffolding

Using a close reference genome (GCA_014131755.1), RagTag was employed to correct misassemblies and bridge contigs into long structural scaffolds:

- **Correction Step:**

  Bash

  pixi run ragtag.py correct GCA_014131755.1_ASM1413175v1_genomic.fna Assembly_step1/contigs.fa
- **Scaffolding Step:**During scaffolding, RagTag can determine the order and orientation of contigs based on their alignment
- **to the reference.**

  Bash

```
ragtag.py scaffold
GCA_014131755.1_ASM1413175v1_genomic.fna
ragtag_output/ragtag.correct.fasta
```

6. Post-Scaffolding Quality Assessment

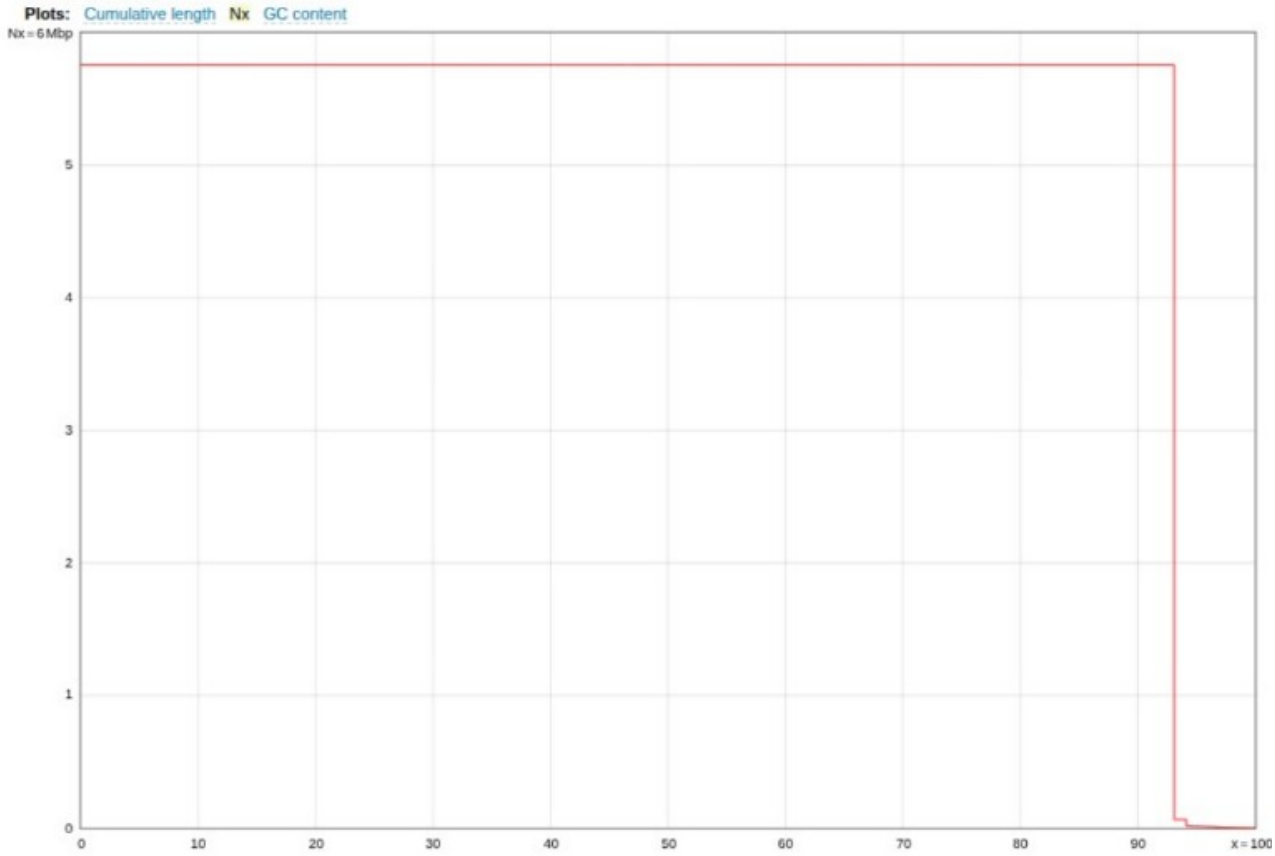
Scaffolded outputs were reassessed using QUAST to measure architectural improvements:

Bash

```
quast.py -o quast_scaffolds ragtag_output/ragtag.scaffold.fasta
```

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Statistics without reference	ragtag.scaffold
# contigs	113
# contigs (>= 0 bp)	535
# contigs (>= 1000 bp)	77
# contigs (>= 5000 bp)	23
# contigs (>= 10000 bp)	12
# contigs (>= 25000 bp)	2
# contigs (>= 50000 bp)	2
Largest contig	5 757 150
Total length	6 182 726
Total length (>= 0 bp)	6 238 610
Total length (>= 1000 bp)	6 158 061
Total length (>= 5000 bp)	6 022 821
Total length (>= 10000 bp)	5 946 032
Total length (>= 25000 bp)	5 820 617
Total length (>= 50000 bp)	5 820 617
N50	5 757 150
N90	5 757 150
auN	5 361 937
L50	1
L90	1
GC (%)	42.95



7. Gene Prediction and Sequence Extraction

Ab initio gene annotation was executed using AUGUSTUS parameterized for prokaryotic structures:

Bash

```
augustus --species=E_coli_K12 ragtag_output/ragtag.scaffold.fasta > scaffold_1.gff
```

Extracted gene coding sequences were subsequently converted into FASTA formatting via GFFread:

Bash

```
gffread scaffold_1.gff -g ragtag_output/ragtag.scaffold.fasta -w multi.fasta
```

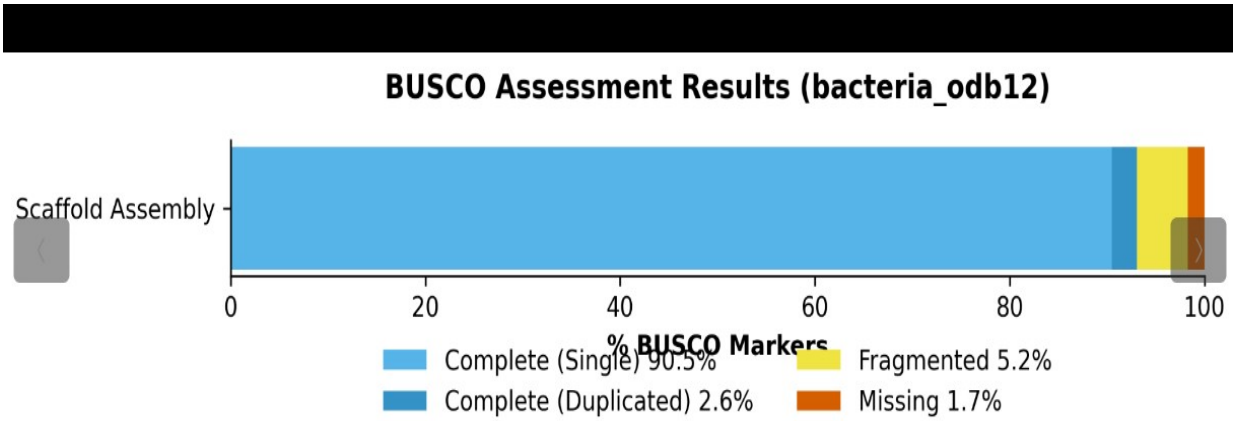
8. Completeness Assessment (BUSCO)

Conserved single-copy ortholog benchmarking was performed using the bacteria_odb12 database:

Bash

busco -i ragtag_output/ragtag.scaffold.fasta -m geno -o busco_output_bact -l bacteria_odb12

- **BUSCO Evaluation:**

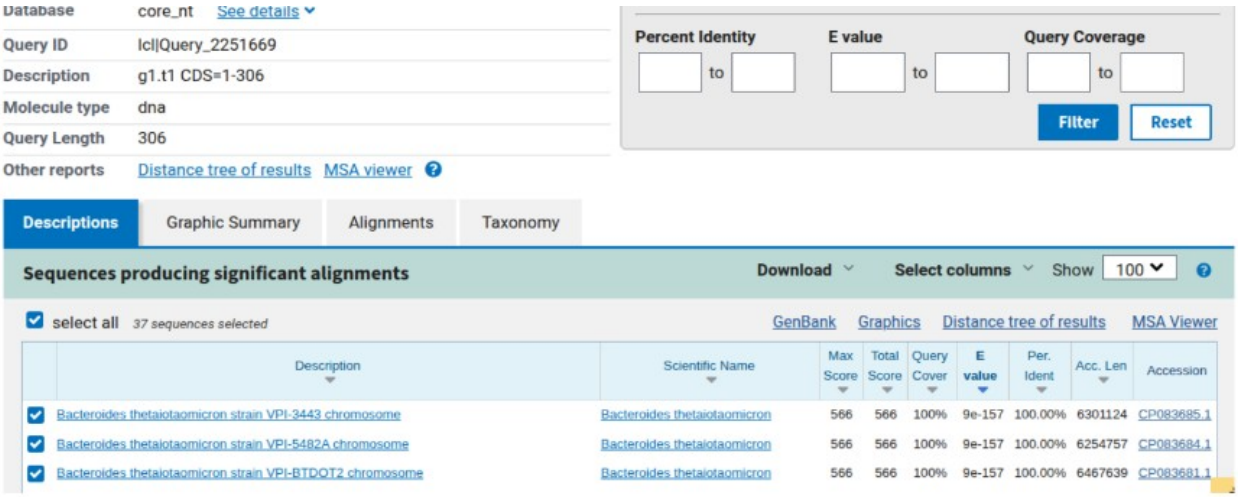


-
  - **Complete (C):** 90.5%%
  - **Fragmented (F):** 5.2%
  - **Missing (M):** 1.7%

- **Interpretation:** The high representation of single copy complete genes (90.5%) alongside minimal fragmentation indicates a complete assembly with high structural integrity.

9. Taxonomic Identification (Remote BLAST)

An extracted sequence from multi.fasta was searched against the NCBI BLAST



- **Identification Result:** *Bacteroides thetaiotaomicron* (100.00% Identity)